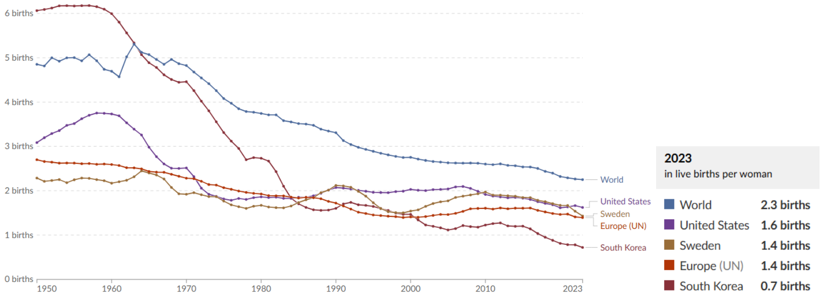
The Distributional Effects of Paid Parental Leave Policies on Fertility and Child Human Capital

Giorgia Conte

February 13, 2026

Motivation: declining fertility across high-income countries

Figure: Fertility rates: births per woman



Data source: UN, World Population Prospects (2024)

Some causes underlying declining fertility

- Parents' trade-off between number of children and investment per child: **Quantity–Quality Trade-Off** (Becker, 1960s–1980s)
- Rising female labor force participation and higher earning: **opportunity cost of time**
- Other theories (e.g. contraception, gender inequality in home production)

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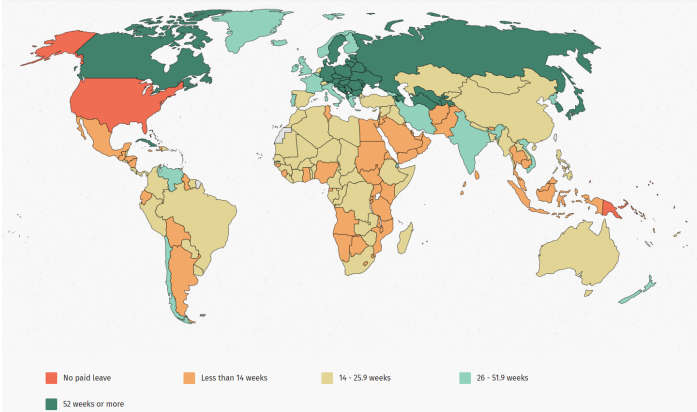
Possible solutions: Policies that reduce the cost of having children

- Childcare subsidies
- Flexible work arrangements
- **Parental leave policies**

Paid leave reserved to mothers across countries



Source: World Policy Center, 2022

Setting

U.S., with declining fertility, lack of support for families, no widespread PPL

This paper

How do paid parental leave (PPL) policies affect fertility?

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  - changes in fertility and labor supply alter parental investment
  - human capital evolves with parental investment throughout childhood
  - key mechanism: quantity-quality trade-off

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Preview of results

Reduced form approach:

- 6 weeks local introduction of PPL $\uparrow$ birth rates (i.e. more births within 1 year), up to 9.6% $\uparrow$ Total Fertility Rate (TFR)
  (average effect across groups of women of different age)

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- 6 extra weeks of PPL, absent fertility responses:
  - Slightly $\uparrow$ investment and child human capital across entire distribution

Contribution to the literature

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Studies on effects of PL reform on Fertility: Lalive et al. (2009), Raute (2019), Dahl (2016), Kleven and Laundais (2024), Conte and Steingrismodottir (2025)
→ take-away: null or positive effects

- New evidence using reforms and US survey data

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- **Main contribution**: unified framework to study these outcomes jointly

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Studies on heterogeneous effects of PPL on fertility: Yamaguci (2019), Erosa et. al (2016), Kim and Yum (2024), Broson et. al (2025) $\rightarrow$ take-away: PPL $\uparrow$ fertility

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Studies on heterogeneous effects of family policies on children's skills formations: Yum (2023), Caucutt et al. (2020), Daruich (2018), Lee et al. (2019), Zhou (2019)
→ take-away: time and money complements ↑ child human capital

- Study effects of **parental leave**

Outline

- Reduced form approach
- Structural model
- Structural estimation of parameters
- Counterfactual analysis

Reduced-form analysis

Overview of reduced-form approach

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- **Estimation strategy**: Diff-in-difference (New Jersey vs Maryland)
  - Outcome: Probability of having given birth in the past 12 months
  - Treatment: women in NJ of fertile age who worked in previous year
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- **Focus sub-sample**: focus on women with "high exposure" to policy
  - Married
  - Employed

Effects on fertility of 2009 introduction of PPL in New Jersey

- Outcome: probability of having given birth in the past 12 months
- Compare New Jersey (T) and Maryland (C), before (Pre) and after (Post) 2009

$$Birth_{icst} = \beta \left(Treat_s \times Post_t \right) + \gamma X_{icst} + \lambda_c + \lambda_t + \lambda_s + \lambda_a + \varepsilon_{icst}.$$

i: individual, s: state, c: county, t: year, a: age, X: ethnicity, education and income

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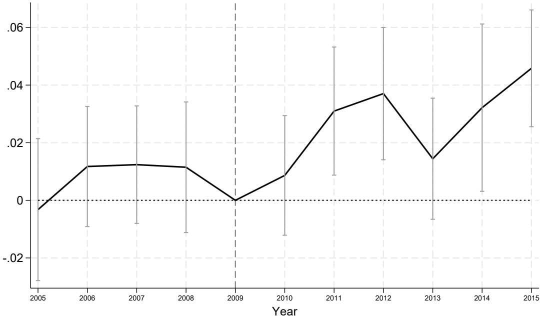

Notes: The sample refers to employed married women with job protection and aged 20–40. 90% CI.

By education

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Robustness checks:

- Release sample restrictions on marriage and employment ▸ More
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- No effect on women with "low-exposure" ▸ low-exposed
- Placebo tests: no effects on ineligible women ▸ Ineligible
- No effect on probability of employment ▸ Employment

Some remaining threats to identification:

- Birth rates vs. completed fertility, could be a temporary effects
- Contamination of Financial Crisis 2007
- Families moving across states

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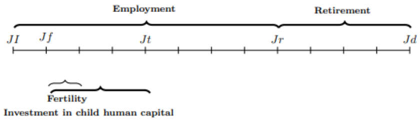
Motivates structural approach

Structural model

Overview of structural approach



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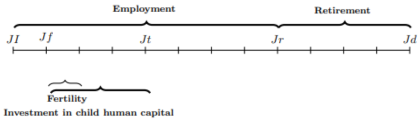
  - Agents enter the life cycle as childless young full-time workers, stay married

Overview of structural approach



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    - infertility risk x, access to partially compensated leave ρ, (stochastic)

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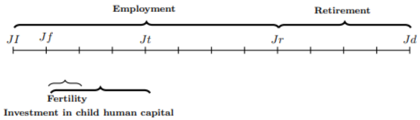

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    - Endogenous fertility (n)
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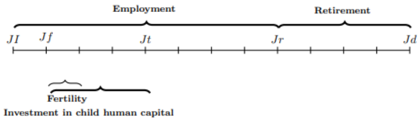

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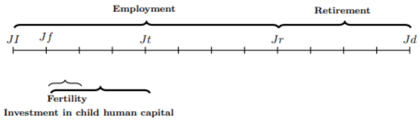

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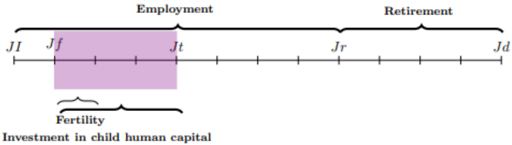
- Counterfactual analysis: simulate NJ policy at national level

Parents' preferences (quantity–quality)

▸ Collective bargaining ▸ Go back

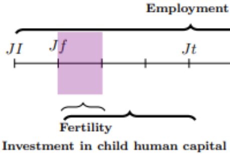


- Household utility:

$$U(c) + \mu U(\ell) + U_k(n, h_k) = \frac{\left(\frac{c}{\Psi(n)}\right)^{\gamma}}{1-\gamma} + \frac{\mu\theta(1-\ell)^{1+v}}{1+v}$$

$$+ \underbrace{\eta_n \frac{n^{\sigma_n}}{\sigma_n}}_{\text{Kids } n} + (n^{\kappa} \underbrace{\eta_{h_k} \frac{h_k^{\sigma_{hk}}}{\sigma_{hk}}}_{\text{Human capital}}) - \underbrace{X_0}_{\text{Fixed cost}}$$

Fertility period's value function and constraints



- Value function:

$$V_j^f(e, z_m, z_f, x, \rho) = \max_{c,n,m,t,\ell,l_f,l_m} u(c) + u(\ell) + U(n, h_k) + \beta E[V_{j+1}^p(e, z'_m, z'_f, n, h_{k_{j+1}})]$$

Fertility choice

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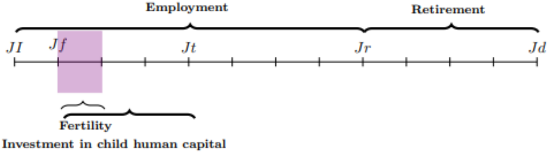

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▸ Fertility choice

- Early childhood human capital ▸ Parental investment ▸ child human capital

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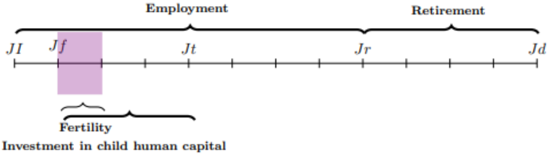

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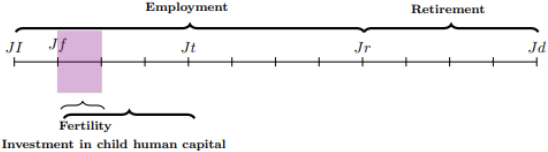

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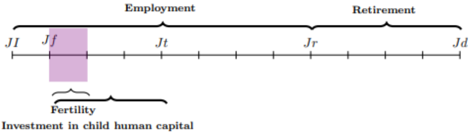
- Household budget constraint:

$$c = (w_m l_m + (w_f(1 - \delta_p(\ell n - \alpha_f t n^{\epsilon}))(1 - \tau) - m(1 - t)n^{\epsilon} + \underbrace{\rho \boldsymbol{Bn}}_{\text{PL compensation}}$$

Parental leave benefits



- Time off-work is unpaid ($\rho = 0$), except for fraction $p_s < 1$ of population entitled to benefits B, ($\rho = 1$), equal to wage replacement $\alpha_s < 1$ for duration d

$$\underbrace{\rho\, B\, n}_{\text{PL compensation}}$$

$$B \;=\; \alpha_s\, w_f\, d$$

$$d = \min\Big(\ell n + \alpha_f\, t_1\, n^{\epsilon},\;\; \bar{T}_s\Big) \qquad\qquad s \in \{\mathscr{B},\, \mathscr{C}\}$$

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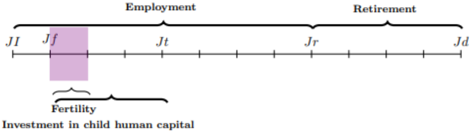

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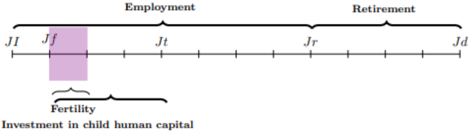

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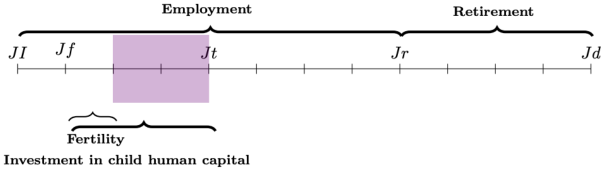
- Baseline scenario: $\alpha_{\mathscr{B}} = 0.85$, $T_{\mathscr{B}} = 6$ weeks, $p_{\mathscr{B}} = 0.4$
- Counterfactual scenario: $\alpha_{\mathscr{C}} = 0.85$, $T_{\mathscr{C}} = +\ 6$ weeks, $p_{\mathscr{C}} = 1$

Following periods' value function and constraints



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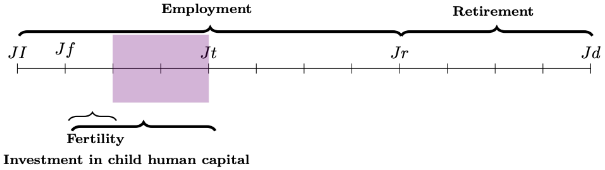

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- Mid-late childhood human capital ▸ Parental investment ▸ child human capital

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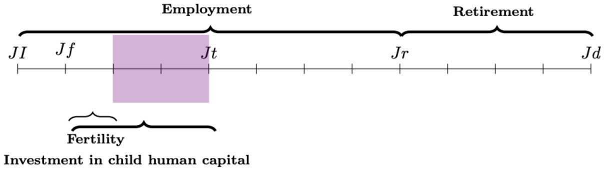

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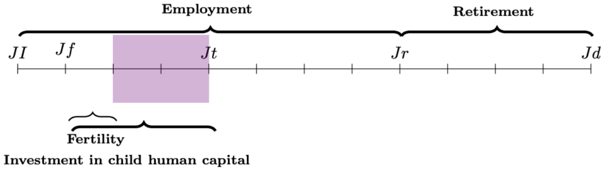
- Time constraints: $l_g = 1 - \alpha_g t n^{\epsilon}, \qquad g \in \{f, m\}$

Following periods' value function and constraints



- Value function:

$$V_j^p(e, z_m, z_f, n, h_k) = \max_{c,m,t,l_f,l_m} u(c) + u(\ell) + U(n, h_k) + \beta E[V_{j+1}^p(e, z_m', z_f', n, h_{k+1})]$$

- Mid-late childhood human capital ▸ Parental investment ▸ child human capital

$$h_{k_{j+1}} = \ell(h_{k_j}, t, m, P), \qquad j \in \{1, 2\}$$

- Time constraints: $l_g = 1 - \alpha_g t n^{\epsilon}, \qquad g \in \{f, m\}$

- Household budget constraint

$$c = (w_f l_f + w_m l_m)(1 - \tau) - m n^{\epsilon}$$

Structural estimation

Data moments and parameters estimation

Data sources:

Panel Study of Income Dynamics (PSID)

- Panel of US household, I use waves from 1993 to 2015
- Information education, income and family structure (including fertility)

Child Development Supplement (CDS)

- Supplementary study on PSID children aged 0 to 18 from 1997 wave.
- Waves 1997, 2002, 2007, 2014 and 2019
  - Time diary for measure on time investment
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Structural estimation of 15 parameters via Simulated Method of Moments (SMM)

- Parameters on parents' preferences and human capital functional form
parameters
- Moments on fertility, child human capital, time spent on childcare and work
moments

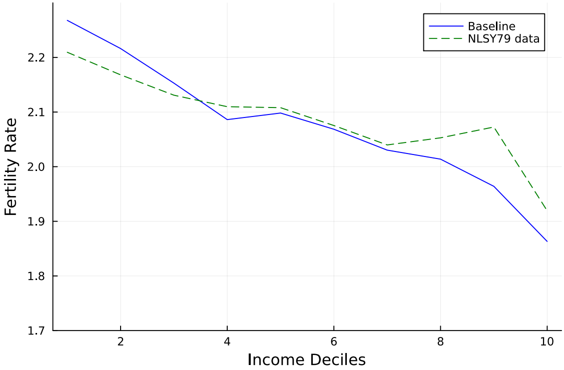
Moments used for SMM calibration

Go back

Moment description	Model	Data
Avg. fertility, college graduates (CG)	1.78	1.72
Avg. fertility, high-school graduates (HS)	1.88	1.90
Share childless, CG	0.13	0.15
Share childless, HS	0.10	0.11
Avg. hk_1, CG	6.64	6.71
Avg. hk_1, HS	5.60	5.82
Avg. hk_2, CG	36.20	36.61
Avg. hk_2, HS	30.53	33.22
Avg. hk_3, CG	48.00	49.20
Avg. hk_3, HS	42.00	42.80
Avg. hours worked, CG with child <5	0.61	0.55
Avg. hours worked, HS with child <5	0.62	0.55
Avg. time investment (t_2), CG	0.18	0.21
Avg. time investment (t_2), HS	0.18	0.20
Avg. time investment (t_3), CG	0.18	0.18
Avg. time investment (t_3), HS	0.18	0.14

Notes: This table reports the moments matched in the simulated–method-of-moments. Fertility and labor supply statistics come from PSID (1993–2015). Human-capital moments refer to families with exactly two children and are computed using CDS data (1997–2019).

Validation: Untargeted Moments Fit



Notes: The figure plots average completed fertility at the intensive margins by maternal wage decile for the baseline model (blue) and NLSY-79 data (green). The sample refers to employed women with children between 40 and 50, reporting positive labor income and living with a partner.

Counterfactual Analysis

Counterfactual Analysis

Policy experiments

- 6 weeks of PPL, 85% wage replacement (New Jersey policy) more

PPL parameters in counterfactual scenario: $\alpha_{\mathscr{B}}$=0.85, $T_{\mathscr{B}}$= +6 weeks, $p_{\mathscr{B}}$=1

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Outcome variables:

- Completed fertility
- Female labor supply
- Parental time investment (per-child)
- Parental monetary investment (per-child)
- Child human capital

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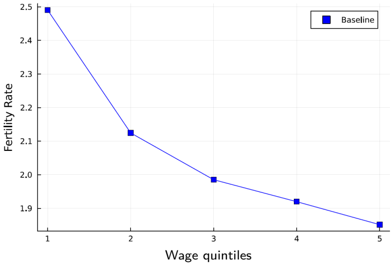
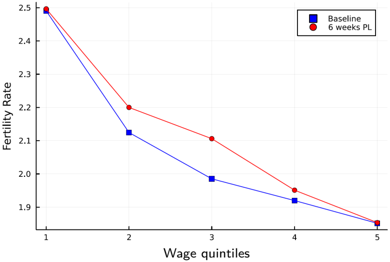
Effects on fertility

Figure: Fertility rates across the wage distribution (intensive margins)



Effects on fertility

Figure: Fertility rates across the wage distribution (intensive margins)



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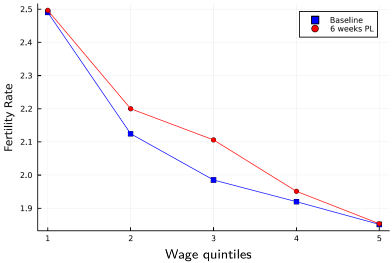
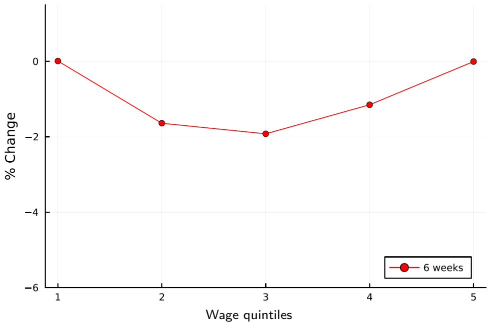


Heterogeneous effects by maternal education:

Policy	HS Graduate	CG Graduate
6 weeks	7%	2%

Effects on female labor supply (FLS)

Figure: Changes (%) in female labor supply under paid leave policies



Changes in FLS in first period of childhood mirror the increases in fertility

▸ more

Effects on female labor supply (FLS)

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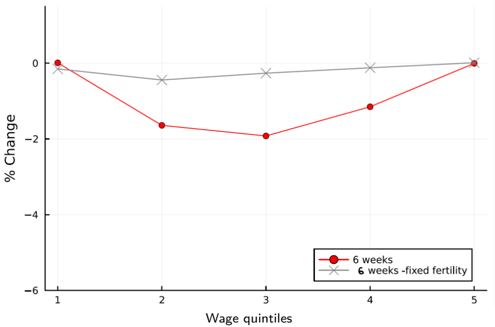
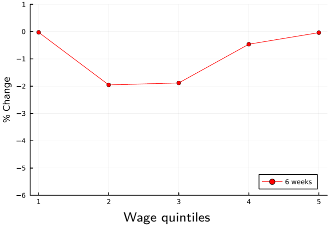


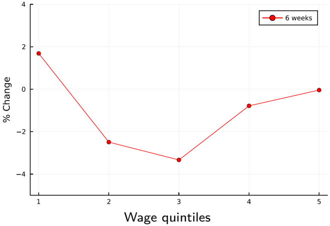
Changes in FLS in first period of childhood mirror the increases in fertility

▶ more

Effects on parental investment

Figure: Changes (%) in time (left) and money (right) investment under PPL





▸ more

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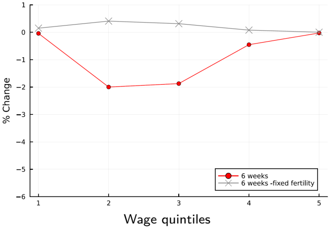
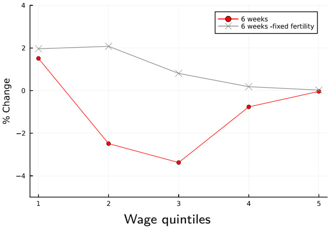
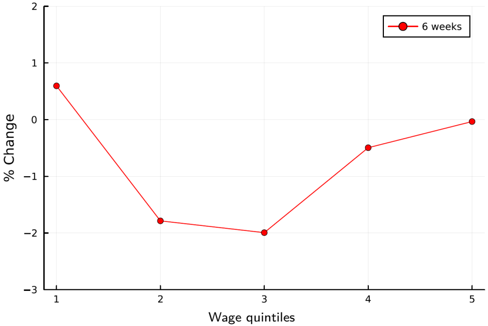




▸ more

Effects on early childhood human capital

Figure: Changes (%) in early childhood human capital under PPL



▸ more

Effects on early childhood human capital

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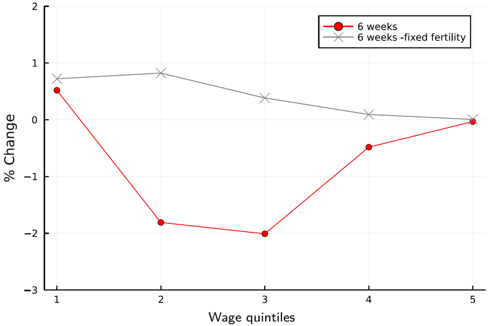
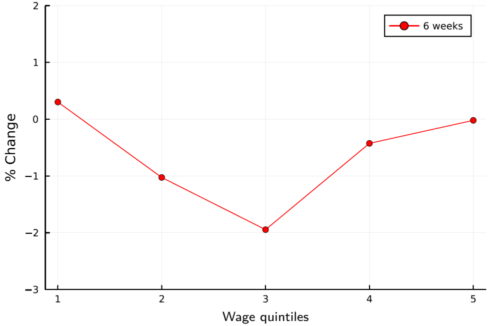


▸ more

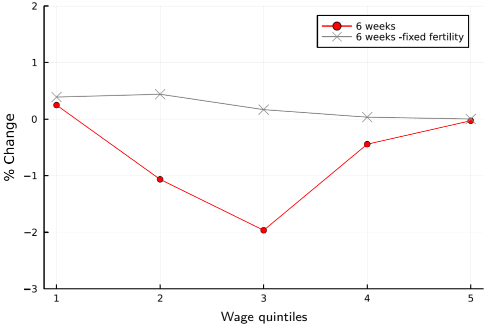
Effects on late childhood human capital

Figure: Changes (%) in late childhood human capital under PPL



Effects on late childhood human capital

Figure: Changes (%) in late childhood human capital under PPL



Conclusions

Main contribution:

First study on the joint effects of PPL on fertility and child human capital

Reduced form approach:

- Empirical analysis suggests that local introduction of PPL ↑ birth probability

Life-cycle model, calibrated to U.S.:

- Introduction at national level ↑ higher order births
- Heterogeneous effect: larger among low-educated and middle-wage families
- Families who increase fertility reduce per-child investment
  quantity-quality trade-off, losses for child human capital
- In absence of fertility responses, child human capital would ↑ slightly

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First study on the joint effects of PPL on fertility and child human capital

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  quantity-quality trade-off, losses for child human capital
- In absence of fertility responses, child human capital would $\uparrow$ slightly

Policy implications:

- Complementary policies to PPL to $\uparrow$ fertility without harming children's development

Research agenda

Model extensions:

- Recalibration to other countries (preferably Sweden!) to analyze the most effective family policy reforms under a fixed budget allocation
- Intra-household bargaining over leave-taking decisions (e.g. effects of "gender equality bonus")
- Extensive margins of FLS to account for job protection dimension, endogenous wage growth through experience accumulation

Other ongoing projects:

- Heterogeneous effects of PL extension on fertility by availability of family support (Quasi-experimental analysis, Denmark)
- Decomposition of causes behind narrowing education-fertility gradient (Denmark)
- Long-term effects of tax reforms combining microsimulation and OLG model

Future projects:

- Dynamic analysis of the interplay between education and fertility decisions
- How AI affects demands for skills (web-scraping of job-search engines)

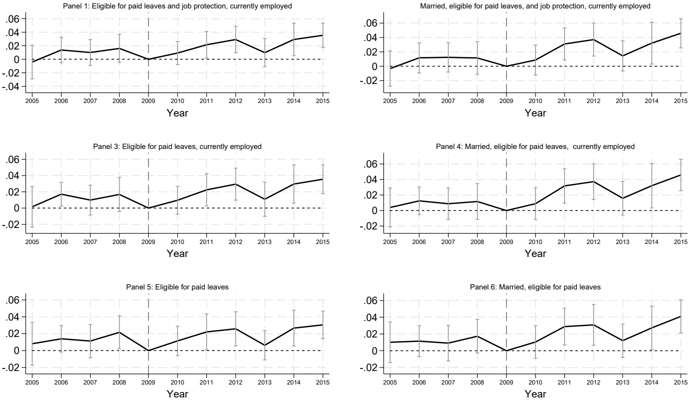
End of Presentation

Thanks for your attention!

conteg@tcd.ie

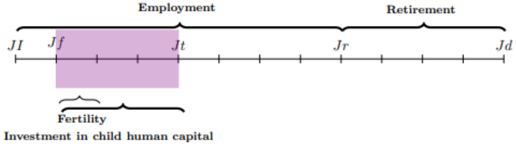
Robustness check: lift current employment and marriage sample restriction

Go back



0.94-1.6 pp ↑ in probability of having a child → 5.8-10.2% ↑ in Total fertility rate (TFR)

Parents' choice



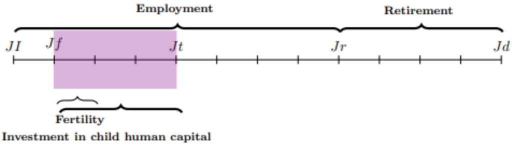
Go back

- Household utility:

$$U(c) + \mu U(\ell) + U_k(n, h_k) = \frac{\left(\frac{c}{\Psi(n)}\right)^{\gamma}}{1-\gamma} + \frac{\theta(1-\ell)^{1+v}}{1+v}$$

$$+ \underbrace{\eta_n \frac{n^{\sigma_n}}{\sigma_n}}_{\text{Kids } n} + n^{\kappa} \underbrace{\eta_{h_k} \frac{h_k^{\sigma_{hk}}}{\sigma_{hk}}}_{\text{Human capital}} - \underbrace{X_0}_{\text{Fixed cost}}$$

Parents' choice



Go back

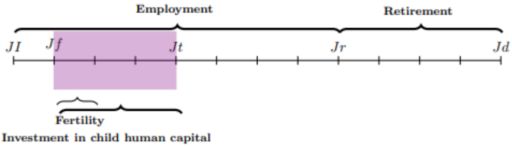
- Household utility:

$$U_f + U_m = \mu[u(c) + u(\ell) + u(n, h_k)] + (1-\mu)[u(c) + u(\ell) + u(n, h_k)]$$

$$U_f + U_m = u(c) + \mu[u(\ell)] + u(n, h_k)$$

$$u(c) + \mu u(\ell) + u(n, h_k) = \frac{\left(\frac{c}{\Psi(n)}\right)^{\gamma}}{1-\gamma} + \frac{\mu\theta(1-\ell)^{1+v}}{1+v} + \underbrace{\eta_n \frac{n^{\sigma_n}}{\sigma_n}}_{\text{Kids } n} + n^{\kappa} \underbrace{\eta_{h_k} \frac{h_k^{\sigma_{hk}}}{\sigma_{hk}}}_{\text{Human capital}} - \underbrace{X_0}_{\text{Fixed cost}}$$

Function of child human capital formation



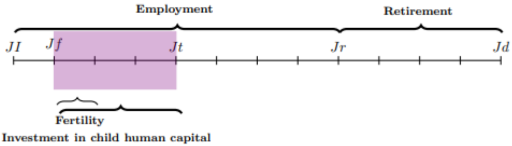
$$h_{k_1} = A_1 \left(\underbrace{\left[\alpha(\delta_e\, t_1)^{\phi_1} + (1-\alpha)(m(1-t_1))^{\phi_1}\right]}_{\text{Parental investment}} \right)^{\frac{1}{\phi_1}}$$

Go back

$$h_{k_{j+1}} = A_{j+1} \left(\underbrace{\gamma {h_{k_j}}^{\rho_j}}_{\text{current human capital}} + \underbrace{(1-\gamma)\left[(\delta_e t_j)^{\phi_2} {m_j}^{1-\phi_2}\right]^{\rho}}_{\text{Parental investment}} \right)^{\frac{1}{\rho}}, \quad j \in \{1,2\}$$

Go back

Function of child human capital formation



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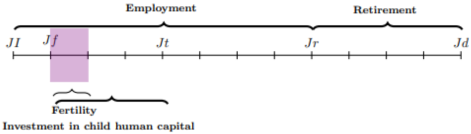
Go back $\{\mu\theta, \alpha, \delta_e, \phi_1, \phi_2, A_{i+1}, \gamma, \rho\} \in P, \quad j \in \{0,1,2\}$

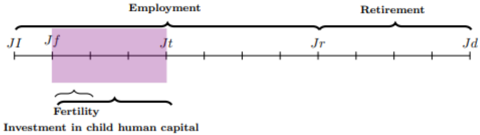
Time allocation during fertility period



- Endogenous Choice of childcare time and leisure
    - Primary caregivers (women) choose fraction of fertility period off-work $(1 - l_f)$
    - Then choose how to split time off-work between leisure ℓ and childcare $\alpha_f t$
    - Maternal childcare time combines time with child during leave and after
    - Motherhood penalty proportional to the time spent out of work, δ_p

Go back

Endogenous parental investment

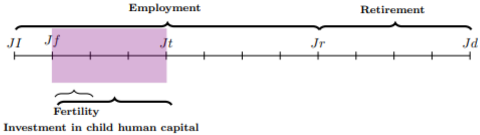


back

- Choice over resources to invest in children
  (money m_j and time t_j) with some economies of scale ϵ

  $\rightarrow$ Child human capital accumulation throughout childhood h_{k_j}

Endogenous parental investment



back

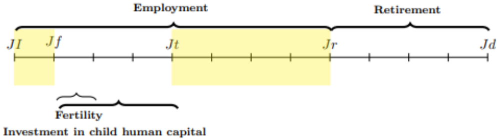
- Choice over resources to invest in children (money m_j and time t_j) with some economies of scale ϵ

  $\rightarrow$ Child human capital accumulation throughout childhood h_{k_j}

  Parental time investment :

  - Maternal and paternal time $\alpha_f t_j + \alpha_m t_j$
  - Allocation of time at work l_g ($g \in \{f, m\}$), and time off-work $1 - l_g$
    in early childhood: childcare and leisure (for women, during leave)
    in mid and late childhood: childcare
  - Productivity of time investment varies with maternal education, δ_e

Workers choice



- Trivial consumption choice with inelastic labor supply ($l = 1$)
- Wage rates given by age, gender, education and productivity: $w(g, e, z, j)$
- Household Value function:

$$V_t(e, z_{t,m}, z_{t,f}) = U(c) + \beta E\{V(e, z_{t+1,m}, z_{t+1,f}, n)\}, J < J_f$$

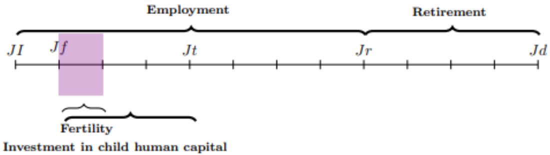
$$V_t(e, z_{t,m}, z_{t,f}) = U(c) + \beta E\{V(e, z_{t+1,m}, z_{t+1,f})\}, J > J_t$$

- Budget Constraint:

$$c = (w_f + w_m)l(1 - \tau)$$

Go back

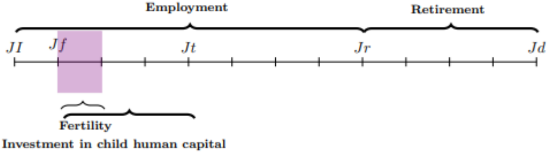
Endogenous fertility



Go back

- Agents enter a single fertility period (choose to have 0-3 children in a 5 year period), n
- Children are born in same model period and are homogeneous

Fertility period's value function and constraints



- Value function:

$$V_j^f(e, z_m, z_f, x, p) = \max_{c,n,m,t,\ell,l_f,l_m} u(c) + u(\ell) + U(n, h_k) + \beta E[V_{j+1}^p(e, z', n, h_{k_{i+1}})]$$

- Time constraints: $l_f = 1 - \ell n - \alpha_f t n^\epsilon$, $l_m = 1 - \alpha_m t n^\epsilon$

- Household budget constraint:

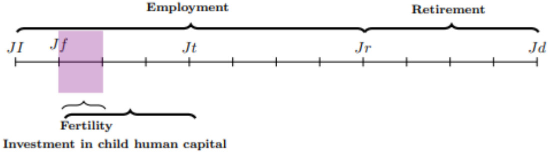
$$c = (w_m l_m + (w_f(1 - \delta_p(\ell n - \alpha_f t n^{\epsilon_2}))(1-\tau) - m(1-t)n^\epsilon + \underbrace{p_s \boldsymbol{B n}}_{\text{PL compensation}}$$

Baseline scenario:	$\alpha_{\mathscr{B}}$=0.85, $T_{\mathscr{B}}$=6 weeks, $p_{\mathscr{B}}$=0.4

- Early childhood human capital

$$h_{k_{i+1}} = \ell(t, m, P), \qquad i = 0.$$

Fertility period's value function and constraints



Go back

- Value function:

$$V_j^f(e, z_m, z_f, x, p) = \max_{c,n,m,t,\ell,l_f,l_m} u(c) + u(\ell) + U(n, h_k) + \beta E[V^p_{j+1}(e, z', n, h_{k_{i+1}})]$$

- Time constraints: $l_f = 1 - \ell n - \alpha_f t n^{\epsilon}$, $l_m = 1 - \alpha_m t n^{\epsilon}$
- Household budget constraint:

$$c = (w_m l_m + (w_f(1 - \delta_p(\ell n - \alpha_f t n^{\epsilon 2}))(1 - \tau) - m(1 - t)n^{\epsilon} + \underbrace{p_s \boldsymbol{B} \boldsymbol{n}}_{\text{PL compensation}}$$

Baseline scenario: $\alpha_{\mathscr{B}}$=0.85, $T_{\mathscr{B}}$=+ 6 weeks, $p_{\mathscr{B}}$=1

- Early childhood human capital

$$h_{k_{i+1}} = f(t, m, P), \qquad i = 0.$$

Parameters Estimation

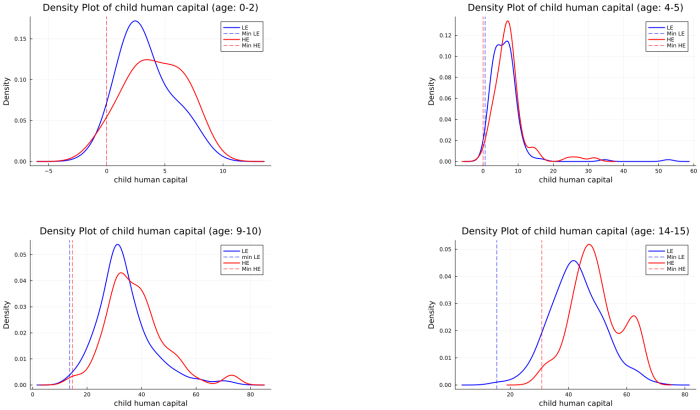
Parameters estimated directly from the data

- PSID: Wage process (age profile and labor productivity)
- CDS-Time Diary:
  - share of time invested by each parent (and joint): α_g
  - proxy of time with each parent while one is PL
- CDS-Child Assessment (Woodcock-Johnson test): children skills
  - Follow Lee and Sheshadri (2015):
    Q is the set of questions $q_i \in (0,1)$ score per question (independent)
  - $Q = \sum(q_i)$ and $\bar{q}$ represents the normalized test score : $\bar{q} = \frac{\sum_{i=1}^{Q} q_i^*}{Q}$
  - Human capital is the (weighted) share of correct answers to the test:
    $h_k = \frac{\bar{q}}{1-\bar{q}}$.

Go back

Go back

Distribution of child human capital by age group and maternal education



Average human capital children in families with 2 children Source: Woodcock-Johnson test. CDS waves 1997, 2002, 2007, 2014, 2019 more

Internally estimated parameters

Go back

- Parameters in preferences and human capital formation function

$$U(c) + U(\ell) + U_k(n, h_k) = \frac{\left(\frac{c}{\Psi(n)}\right)^{\gamma}}{1-\gamma} + \frac{\mu\theta(1-\ell)^{1+v}}{1+v} + \underbrace{\eta_n \frac{n^{\sigma_n}}{\sigma_n}}_{\text{Kids } n} + n^{\kappa} \underbrace{\eta_{h_k} \frac{h_k^{\sigma_{hk}}}{\sigma_{hk}}}_{\text{Human capital}} - \underbrace{X_0}_{\text{Fixed cost}}$$

- Moments: fertility, children skills and time spent with parents

$$h_{k_1} = \left(\underbrace{A_1 \left[\alpha(\delta_e\, t_1)^{\phi_1} + (1-\alpha) m (1-t_1)^{\phi_1} \right]}_{\text{Parental investment}} \right)^{\frac{1}{\phi_1}}$$

$$h_{k_{i+1}} = \left(\underbrace{\gamma h_{k_i}{}^{\rho}}_{\text{current human capital}} + \underbrace{(1-\gamma) A_i \left[(\delta_e t_i)^{\phi_2} m_i{}^{1-\phi_2} \right]^{\rho}}_{\text{Parental investment}} \right)^{\frac{1}{\rho}}, i \in \{1, 2\}$$

Internally estimated parameters

Go back

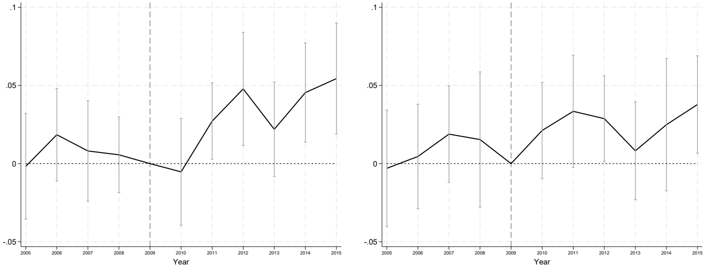
Parameter	Value	Description
η_n	1.15	Utility children's n
σ_n	0.21	Utility children's n
κ	0.13	Utility children n
X_{e1}	2.6	Fixed cost, HS
X_{e2}	2.9	Fixed cost, CG
η_{h_k}	0.2	Utility children's h_k
σ_{hk}	0.49	Utility children's h_k
ϕ_1	-0.6	Elasticity p. (h_{k_2}, h_{k_3})
ϕ_2	0.38	Elasticity p.
α	0.25	Relative weight of time vs money in h_{k_1}
A_{I1}	2.55	Productivity inv. (h_{k_1})
A_{I2}	8.4	Productivity inv. (h_{k_2})
A_{I3}	4.0	Productivity inv. (h_{k_3})
δ_{CG}	1.16	Productivity time inv. CG vs HS (h_{k_3})
$\mu\theta$	1.66	Disutility of work

Notes: This table reports the parameter values resulting from the SMM estimation.

Differences by maternal education

TWFE estimates on probability of giving birth

(A) Women with less than college degree (B) Women with colege degree or higher



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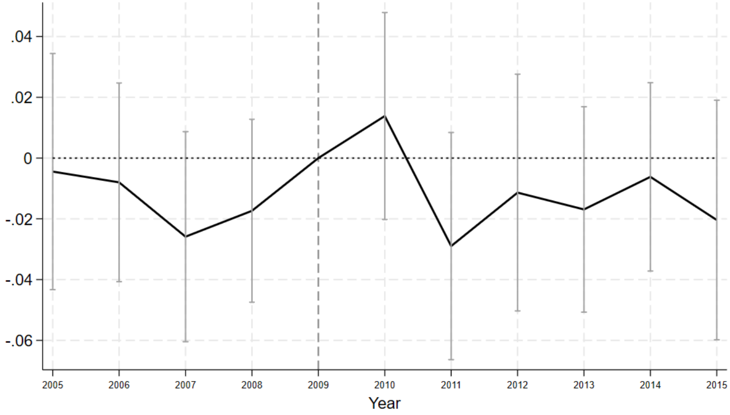
Effects on Employment



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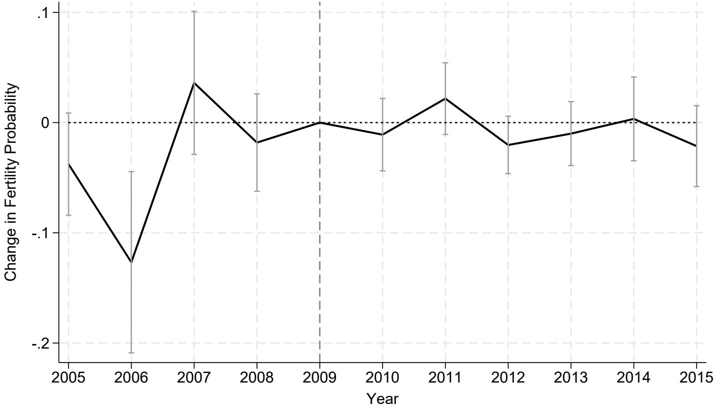
Effects on women in least favorable conditions: single and currently unemployed



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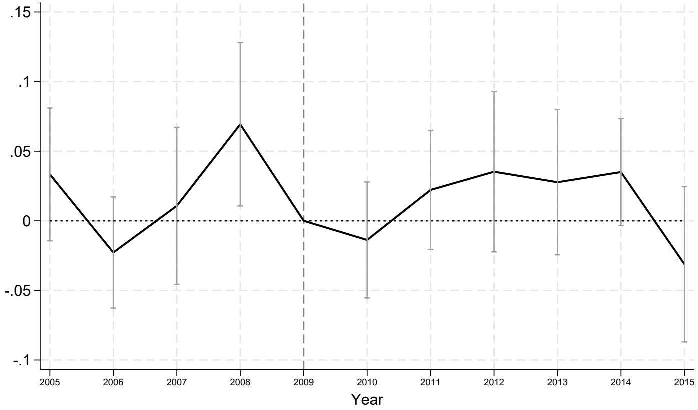
Placebo: effects on married but ineligible women



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Effects on Female Labor Supply

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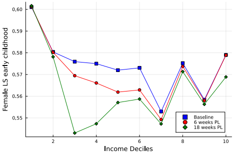
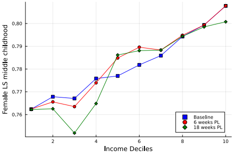
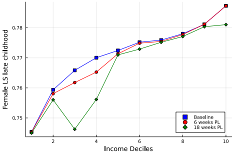
Figure: Female labor supply under paid leave policies



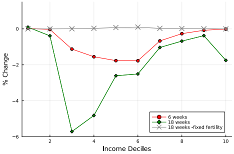
Panel A: FLS in early childhood



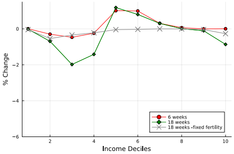
Panel B: FLS in mid childhood



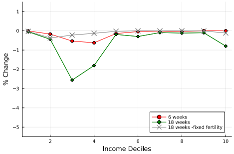
Panel C: FLS in late childhood



Panel D: % changes FLS in early childhood



Panel E: % changes FLS in middle childhood



Panel F: % changes in FLS in late childhood

Effects on child human capital

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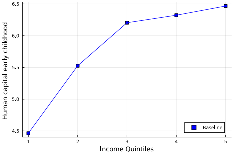
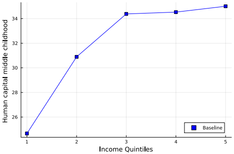
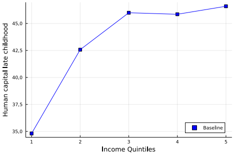
Figure: Child human capital by stage of childhood and policy scenario



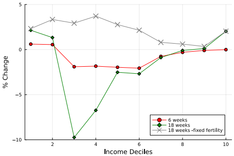
Panel A: Child human capital in early childhood



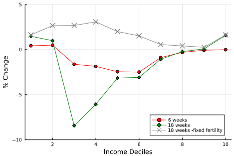
Panel B: Child human capital in middle childhood



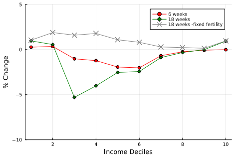
Panel C: Child human capital in late childhood



Panel D: % change child human capital in early childhood



Panel E: % change child human capital in middle childhood



Panel F: % change child human capital in late childhood

Effects on monetary investment

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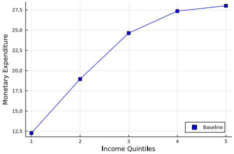
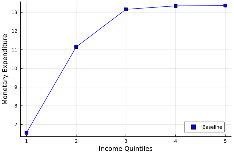
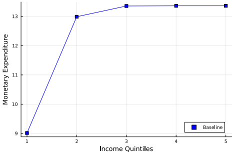
Figure: Monetary investment in children by stage of childhood and policy scenario



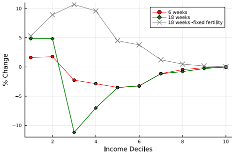
Panel A: Money inv. in early childhood



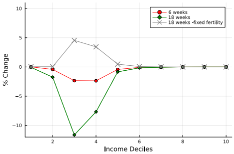
Panel B: Money inv. in middle childhood



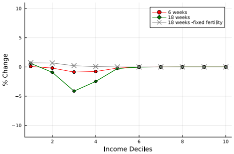
Panel C: Money inv. in late childhood



Panel D: % change money inv. in early childhood



Panel E: % change money inv. in middle childhood



Panel F: % change money inv. in late childhood

Descriptive statistics

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Table: Weighted Means by Treatment Status (Maryland vs. New Jersey) and Differences

Variable	Control (Maryland)	Treatment (New Jersey)	Difference (C − T)
Age	31.698 (0.054)	32.293 (0.043)	-0.595^{***}
Dummy for white ethnic group	0.654 (0.005)	0.696 (0.004)	-0.042^{***}
Education (years)	8.256 (0.023)	8.239 (0.020)	0.017
Dummy for married	0.801 (0.004)	0.828 (0.004)	-0.027^{***}
Family income	103,000 (662.906)	110,000 (659.023)	$-7{,}036.743^{***}$
Number of children	1.243 (0.012)	1.252 (0.010)	−0.009
Employed	0.928 (0.003)	0.922 (0.002)	0.005
Wage income	39,246 (302.512)	40,751 (293.107)	$-1{,}504.972^{***}$
Weeks worked	48.141 (0.097)	48.050 (0.081)	0.090
Family size	3.425 (0.014)	3.446 (0.012)	−0.021
Age youngest child	36.704 (0.451)	36.383 (0.392)	0.322
Dummy for US citizen	0.885 (0.003)	0.835 (0.003)	0.050^{***}

Notes: Entries are weighted means with robust standard errors in parentheses. Differences are Control minus Treatment (Maryland − New Jersey). Asterisks denote significance levels: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Sample sizes: Maryland $N = 14{,}088$; New Jersey $N = 19{,}388$.

Raw fertility rates

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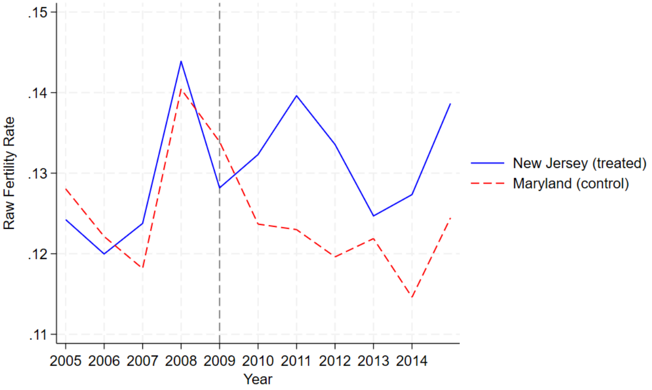

Choice of Control

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Table: Differences in weighted means (Control State $-$ New Jersey)

Variable	Maryland	Pennsylvania	Virginia	Ohio	Illinois	Delaware	Connecticut	New York
Age	-0.595^{***}	-0.798^{***}	-0.983^{***}	-1.162^{***}	-0.956^{***}	-0.844^{***}	-0.254^{***}	-0.523^{***}
Dummy white group	-0.042^{***}	0.186^{***}	0.052^{***}	0.197^{***}	0.079^{***}	0.059^{***}	0.102^{***}	0.017
Education (years)	0.017	-0.220^{***}	-0.157^{***}	-0.421^{***}	-0.205^{***}	-0.369^{***}	0.002	0.035^{***}
Dummy for married	-0.027^{***}	-0.036^{***}	0.005	-0.033^{***}	-0.020^{***}	-0.046^{***}	-0.041^{***}	-8418^{***}
Family income	$-7{,}036^{***}$	$-29{,}000^{***}$	$-16{,}900^{***}$	$-35{,}600^{***}$	$-20{,}900^{***}$	$-23{,}000^{***}$	$-5{,}301^{***}$	-0.042^{***}
Number of children	-0.009	0.027^{**}	-0.050^{***}	0.128^{***}	0.078^{***}	0.046	-0.051^{***}	-0.047^{***}
Employed	0.005	-0.003	-0.005	-0.009^{***}	-0.008^{***}	0.005	0.003	-0.000
Wage income	$-1{,}505^{***}$	$-10{,}100^{***}$	$-5{,}883^{***}$	$-12{,}400^{***}$	$-8{,}014^{***}$	$-6{,}464^{***}$	$-2{,}109^{***}$	-2756^{***}
Weeks worked	0.090	-0.044	0.070	0.006	-0.118	0.188	0.212	0.058
Family size	-0.021	-0.069^{***}	-0.108^{***}	0.023	0.044^{***}	-0.028	-0.102^{***}	-0.014
Age youngest child	0.322	-0.415	1.812^{***}	-2.828^{***}	-0.918^{*}	-0.080	1.698^{**}	2.916^{***}
Dummy US citizen	0.050^{***}	0.123^{***}	0.074^{***}	0.139^{***}	0.055^{***}	0.088^{***}	0.047^{***}	0.027^{***}

Notes: Entries are mean differences (Control $-$ Treatment), where the treatment state is New Jersey, captured until 2009. Asterisks denote significance levels: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Variables match the original specification. Monetary differences are in dollars.